

Charotar University of Science and Technology ,CHANGA
Fourth Semester of B.Tech.(EC) Theory Exam
November 2013

Subject: EC209 Microprocessors & Microcontroller

Date: 12/11/2013 (Tuesday) Time: 01:30p.m. to 04:30 p.m. Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to the right indicate *full* marks.

(4) Rough work is done in the last page of main answer book, don't write anything on Question paper.

(5) Make suitable assumptions and draw neat figures wherever if required.

(6) Section I and II must be attempted in separate answer sheets.

Section - I

Q.1 With neat sketch explain architecture of 8085 in detail. **[07]**

Q.2 (a) 1. Explain the difference between the machine language and the assembly language of the 8085 microprocessor. **[03]**

2. Define: **[04]**
i. machine cycle
ii. Instruction cycle
iii. Microprocessor.
iv. Opcode

Q-2(b) Draw and explain block diagram of 8255. **[06]**

OR

Q.2(b) List features of 8251. Draw and explain block diagram of 8251. **[06]**

Q.3 Attempt any three. **[15]**

1. With neat sketch explain programming model of 8085.
2. With help of neat sketch explain handshake mode of 8255.
3. Define PSW. Explain each field of it with suitable example.
4. Interface 4-KB of memory with 8085 with address range A000H-AFFFFH.
5. List software tools are used for programming of microprocessor.

Section - II

Q.4 Draw and explain internal architecture of 8051 microcontroller. [07]

Q.5(a) Write an assembly program for 8051 to transmit "CHARUSAT" serially with baud rate 9600. Assume crystal frequency = 11.0592 MHz [07]

Q.5(b) Interface LED with P1.0. Write an assembly program to toggle LED every one second. [06]

OR

Q.5(b) With neat diagram explain interrupt system of 8051 along with IE register of 8051. [06]

Q.6 Attempt any three. [15]

1. Classify instruction set of 8085.
2. Write a program for 8051 to multiply two 8-bit number without using MUL instruction.
3. With help of suitable example explain bit manipulated instruction of 8051.
4. Explain modes of 8051 data communication.
5. Specify the register contents and the flag status as the following instructions are executed.

A	B	C	S	Z	CY
XX	XX	XX	0	0	0

MVI A,FFH

MVI B,FFH

ADD B

MOV C,FFH

ADC C

HLT
